



UNIVERSITI TEKNOLOGI MALAYSIA

PROJECT 1

SYSTEM ANALYSIS AND DESIGN

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1.0 Introduction

In this project, we are excited to collaborate with Lembaga Kemajuan Pertanian Kemubu (KADA) to develop an online registration and loan application system. Currently, all of KADA's procedures are manual, where the applicants must complete the form and submit it in person. This manual method takes a lot of time and is again easy to make mistakes. The clerks need to manually enter data and handle applications, resulting in inefficiencies and delays.

We propose to digitize these processes by creating a computerized system that allows users to complete their registration and loan applications online. The new system will feature a user-friendly interface where candidates will be able to electronically fill out forms, upload documents, and enter their personal information. Authorized KADA staff will have efficient access to this securely stored data in a centralized database for review and processing. The efficient review process, automated notifications, and application status tracking features of the loan application system would greatly shorten the time needed for approvals and increase applicant satisfaction.

The implementation of this online system will bring numerous benefits to KADA and its members. It will minimize human data input errors, cut down on paperwork, and improve operational efficiency. With faster processing times and more flexibility, the new system will increase member satisfaction and perhaps get more people to apply for loans and join the cooperative. This transformation will not only simplify administrative processes but also contribute to the growth and success of KADA.

2.0 Background of study

The current system, or the as-is system in KADA, is a manual system, where we saw that it has a very low efficiency and productivity. According to the list below, are the services that KADA provided to their staff currently.

1. Registration to become a membership of KADA

In the current system, all the registration procedures have to undergo a manual process, such as filling in the paper form. For example, users must pick up the form in person at the headquarters in order to fill it in. In the registration process, most of the personal information of the users that are private and sensitive will be collected. Then, the users have to submit it manually also by submitting it at the headquarters. After the filled in and submitting process is done, then the registration paper would be further reviewed by the staff itself manually whether the users are qualified to become their member or not.

2. Applying a loan in KADA

Same as the registration process to become a membership of KADA, applying for a loan is also by filling in the paper form manually. The difference is in applying for a loan form, it would ask more detail about the user income information and guarantor. After the users have submitted it into the headquarters, it would be reviewed by the staff and the approval would be given by the higher position. The approval is given by conducting a monthly meeting between the staff and the higher position.

The current system has a lot of disadvantages, that is why we proposed a solution by making all these services online. By having it online, it could save a lot of time for the users and KADA, and it could also increase the KADA work efficiency.

2.1 Method of information gathering

We have determined that the interactive way of information collecting for this project will be conducting interviews. This approach works well since it allows us to get the essential data for the system requirements. In order to avoid asking the interviewee the same questions twice, we have worked with other groups to create a set of questions in advance of the interview.

There are 2 types of questions which are open-ended questions and close-ended questions. The purpose of open-ended questions is to learn about the primary preferences and system objectives of the clients. Respondents are able to provide specific details about their requirements and expectations by answering these questions. Closed questions are created to gather precise data, trustworthy information and help us make informed decisions with clear data. By combining the two kinds of inquiries, we are able to fully comprehend the needs of our consumers.

Following our preparation, on May 14, 2024, we conducted an online interview with our client, Encik. Ahmad Rohailan Bin Hani dan Puan. Nor Zamani Binti Ismail from Lembaga Kemajuan Pertanian Kemubu. During this conversation, we were able to discover more about the system requirements of the customer. The customer also supplied multiple comprehensive explanations of the system requirements. Lastly, the client discussed their needs for the proposed system as well as the functioning of the current system.

2.2 Summary from method used (include example Interview/Questionnaire/Observation)

 
KOPERASI KAKITANGAN KADA KELANTAN BERHAD

PERMOHONAN MENJADI ANGGOTA

MAKLUMAT PEMOHON

NAMA		JANTINA	☐	☐	LELAKI
NO. KP					PEREMPUAN
TARAF PERKAHWINAN					
ALAMAT RUMAH		AGAMA	☐	☐	ISLAM
					LAIN-LAIN
					Nyatakan
POSKOD					
NEGERI		BANGSA	☐	☐	MELAYU
NO. ANGGOTA					LAIN-LAIN
NO. PF					Nyatakan
JAWATAN & GRED					
ALAMAT PEJABAT					
(Tempat Bertugas)					
POSKOD					
NO. TEL/FAX					
NO. TEL BIMBIT					
NO. TEL RUMAH					
GAJI BULANAN	RM				

MAKLUMAT KELUARGA DAN PEWARIS

BIL	HUBUNGAN	NAMA	NO. K/P @ NO. SRT BERANAK

* SILA BUAT LAMPIRAN JIKA RUANGAN YANG DISEDIAKAN TIDAK MENCUKUPI

YURAN DAN SUMBANGAN

JIKA DI TERIMA SEBAGAI ANGGOTA, SAYA BERSETUJU MEMBAYAR YURAN DAN SUMBANGAN BULANAN SEPERTI DI BAWAH :

BIL	PERKARA	RM
1	FEE MASUK	
2	MODAH SYER *	
3	MODAL YURAN	
4	WANG DEPOSIT ANGGOTA	
5	SUMBANGAN TABUNG KEBAJIKAN (AL-ABRAR)	
6	SIMPANAN TETAP	
7	LAIN-LAIN	

* MINIMA MODAL SYER ADALAH SEBANYAK RM 300.00 DAN TIDAK MELEBIHI 1/3 DARIPADA MODAL SYER KOPERASI DAN HENDAKLAH DIELASKAN DALAM TEMPOH 6 BULAN DARI TARIKH KELULUSAN MENJADI ANGGOTA.

Figure 2.2.1 Registration Form KADA

Figure 2.2.1 presents a comprehensive overview of the KADA organization membership registration form. Aspiring members must complete this form to join the esteemed organization. The registration process begins with applicants providing their personal details, including their full name, residential address, identification number, and other relevant information. Following this, applicants are required to furnish details about their family. The final section of the form is reserved for KADA's administrative use, where the organization will document the membership fee, which is set at 50 Ringgit per month.

KOPERASI KAKITANGAN KADA KELANTAN BERHAD

BUTIR-BUTIR PENJAMIN												
PENJAMIN 1		PENJAMIN 2										
NAMA	_____	NAMA	_____									
NO. KP	_____	NO. KP	_____									
NO. ANGGOTA	_____	NO. ANGGOTA	_____									
Adalah dengan ini saya/kami dengan rela hati bersetuju bersama-sama atau berasingan bertanggungjawab sepenuhnya ke atas ansuran harga barangan atau pembiayaan ini. Kami berjanji akan membayar balik kesemua ansuran hutang yang diberikan kepada peminjam di atas jika sekiranya beliau tidak dapat membayar balik ansuran itu mengikut perjanjian yang telah dipersetujui.												
	SAKSI 1		SAKSI 2									
Tandatangan Penjamin 1	No.K/P :	Tandatangan Penjamin 2	No.K/P :									
Bertarih :	No. Telefon :	Bertarih :	No. Telefon :									
PENGESAHAN MAJIKAN												
Kami mengesahkan bahawa : _____ No. K/P : _____ telah memberikan butir-butir peribadi dan maklumat pendapatan selaras dengan rekod pekerjaan kakitangan tersebut. Kami juga mengesahkan bahawa kakitangan adalah berjawatan tetap.												
Gaji pokok sebulan	RM		COP									
Gaji bersih sebulan	RM											
		Tandatangan Sah/Cop	Bertarih :									
KEGUNAAN PEJABAT												
Tarikh Permohonan diterima Jumlah Syer dan Yuran Terkumpul Jumlah baki pinjaman sedia ada : ☐ AL - BAI ☐ AL-INAH ☐ SKIM KHAS ☐ CUKAI JALAN		Bil. Permohonan ☐ BAIK PULIH KENDERAAN ☐ KARNIVAL MUSIM ISTIMEWA ☐ LAIN-LAIN (Nyatakan)										
Jumlah Potongan Koperasi Nota Akaun : <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th>Debit</th> <th>Kredit</th> <th>Ansuran</th> </tr> </thead> <tbody> <tr> <td>1 Belian</td> <td>Bank</td> <td></td> </tr> <tr> <td>2 Si berhutang</td> <td>Jualan</td> <td></td> </tr> </tbody> </table>		Debit	Kredit	Ansuran	1 Belian	Bank		2 Si berhutang	Jualan		Tandatangan Pengurus Besar Bertarih :	
Debit	Kredit	Ansuran										
1 Belian	Bank											
2 Si berhutang	Jualan											
Keputusan Mesyuarat Jawatan Kuasa Perniagaan ☐ DILULUSKAN Bilangan Mesyuarat Tarikh		Catatan Tandatangan Pengerusi I/K Perniagaan Bertarih :										

Figure 2.2.3 Loan Application Form

Figure 2.2.2 and figure 2.2.3 are the overview of the loan application form. The applicator must be a staff of the KADA In order the to apply loan. The applicator need to fill in the loan requirement such the type of loan and how much the applicant wants to borrow. Next, the applicant need to fill in the personal information and also the applicant's statement. After that, the applicator need to have 2 guarantor in order to apply this loan. Lastly , the part of the verification from the employer and the office use will be fill by the KADA staff.



NAMA: MOHD ROSLAN B ISMAIL

NO. K/P: 740807-03-5479 **NO. PF:** 2273

NO. AHLI:
1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM	RM	RM	RM	RM

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

**PENGESAHAN BAGI PENYATA KEWANGAN
AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023**

Saya **MOHD ROSLAN B ISMAIL** No. Ahli: **1579** mengesahkan bahawa
Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: _____
Tarikh: _____

☐ Setuju

☐ Tidak Setuju.....
 (Nyatakan sebabnya)

Nota:

1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Figure 2.2.4 example of Cooperative Member Financial Report

Figure 2.2.4 showcases a detailed example of the Monthly Cooperative Member Financial Report. This comprehensive report provides vital information regarding the applicant's stock holdings, including share capital and other relevant financial data. It also outlines the details of any loans the applicant has taken, categorized by the type of loan, such as Al-Bai, Al-Innah, among others. The final section of the report is dedicated to the applicant's verification of the financial statement, ensuring accuracy and transparency in the financial reporting process.



JADUAL PEMBAYARAN BALIK PEMBIAYAAN SKIM AL-BAI'UBITHAMAN AAJIL/BAI AL-INAH

KADAR KEUNTUNGAN	4.20%					
TEMPOH (TAHUN)	1	2	3	4	5	6
TEMPOH (BULANAN)	12	24	36	48	60	72
JUMLAH PEMBIAYAAN	ANSURAN BULANAN					
1,000.00	86.83	45.17	31.28	24.33	20.17	17.39
2,000.00	173.67	90.33	62.56	48.67	40.33	34.78
3,000.00	260.50	135.50	93.83	73.00	60.50	52.17
4,000.00	347.33	180.67	125.11	97.33	80.67	69.56
5,000.00	434.17	225.83	156.39	121.67	100.83	86.94
6,000.00	521.00	271.00	187.67	146.00	121.00	104.33
7,000.00	607.83	316.17	218.94	170.33	141.17	121.72
8,000.00	694.67	361.33	250.22	194.67	161.33	139.11
9,000.00	781.50	406.50	281.50	219.00	181.50	156.50
10,000.00	868.33	451.67	312.78	243.33	201.67	173.89
11,000.00	955.17	496.83	344.06	267.67	221.83	191.28
12,000.00	1042.00	542.00	375.33	292.00	242.00	208.67
13,000.00	1128.83	587.17	406.61	316.33	262.17	226.06
14,000.00	1215.67	632.33	437.89	340.67	282.33	243.44
15,000.00	1302.50	677.50	469.17	365.00	302.50	260.83
16,000.00	1389.33	722.67	500.44	389.33	322.67	278.22
17,000.00	1476.17	767.83	531.72	413.67	342.83	295.61
18,000.00	1563.00	813.00	563.00	438.00	363.00	313.00
19,000.00	1649.83	858.17	594.28	462.33	383.17	275.06
20,000.00	1736.67	903.33	625.56	486.67	403.33	347.78

Dikemaskini 10 Mei 2023

Mesyuarat Lembaga bil 3/2023 7 Mei 2023

Figure 2.2.5 Loan Repayment Schedule

Figure 2.2.5 presents the loan repayment schedule in a detailed tabular format. This table illustrates a fixed profit rate of 4.20% and offers repayment durations ranging from 1 to 6 years. Each year corresponds to a specific monthly installment amount. For instance, if the loan amount is RM 1,000 with a repayment period of 1 year, the applicant is required to pay RM 86.83 per month, in addition to the RM 50 monthly membership fee. This schedule provides a clear and structured overview of the repayment obligations for the applicants.

Question for the interview

1. Bolehkah pihak tuan memberikan penjelasan mengenai sistem yang sedia ada?
2. Apakah kriteria khusus yang diambil kira oleh pihak koperasi dalam proses kelulusan pinjaman, dan berapa lama biasanya masa yang diambil dari permohonan sehingga kelulusan pinjaman diberikan?
3. Bahagian manakah yang tuan rasa perlu dikekalkan seperti yang sedia ada?
4. Jika permohonan untuk menjadi ahli koperasi tidak diluluskan, adakah boleh untuk membuat rayuan semula?
5. Adakah terdapat jumlah maksimum pinjaman yang boleh kita pinjam?

3.0 Problem statement

1. Loss of information

Most of the records are stored in paper such as the registration documents and also the loan documents. Paper documents can be easily damaged by water, fire or mishandling. Once damaged, the information on them may become unreadable or entirely lost. Other than that, paper documents can be misplaced and lost due to human error.

2. Accumulation of data when access or update

The clerk must locate the paper back in order to update the data, which is a difficult operation because they must scan and identify each paper in their safe in order to locate the particular one. As a result, it will take longer to finish a single task—updating the user's information.

3. Highly dependent on human

Using paper forms greatly reduces productivity and efficiency because handling, filing, and finding these documents take a lot of time and effort. The dependence on paper slows down operations, causes delays, and keeps employees from focusing on more important tasks, ultimately making the organization less productive.

4. Difficulty in distributing and receiving the application

In order to obtain the form and submit it, applicants must make an extensive journey to the headquarters. They are certain to be burdened by this since they will need to find time to travel to the headquarters and spend money on transportation.

4.0 Proposed solution

The system will serve as a cutting-edge substitute for the outdated paperwork system.

Thus, the system will perform the same basic tasks as the original paperwork system, including handling the registration form for the new member and also the registration form to apply for a loan for the member of the organization. On the other hand, The shift from physical documentation to an online form is the main upgrade for the system. In the past, the organization primarily employed paper in their operations to store user information and to verify users during the loan application and registration processes. Initially, the current system only had three primary user views: the applicator, Ahli Lembaga Pengarah (ALK), and the clerk. However, after a few discussions, we concluded that the admin roles would be essential for the new online system.

Firstly, from the view of the applicants, The new system will definitely be beneficial for the user as the online system offers many benefits for the applicants. With the current paper system, people have to travel to the headquarters to get and submit forms, which takes a lot of time and money. This can especially be difficult for those who live far away or have busy schedules. In contrast, the online system allows people to access and submit forms anytime and from anywhere with an internet connection. This means they don't need to make a special trip, saving both time and travel costs. The online system is available 24/7, making it much more convenient. It also provides instant confirmation when an application is submitted, so applicants don't have to worry whether their forms are received by KADA or not. Additionally, online forms often help prevent mistakes by checking for missing or incorrect information before submission. Overall, the online application system makes the process faster, easier, and cheaper for everyone involved.

Secondly, for the members of the Board of Directors (ALK), their job of checking membership and loan applications becomes much easier with the new online system. Before, this process was slow and difficult because they had to be at the office and deal with lots of paperwork, which limited their flexibility and caused delayed decisions. With the online system, ALK members can do their work from anywhere and at any time, without needing to travel or handle physical documents. This means they can approve applications faster and more easily. The system also provides instant updates and notifications, making their work more efficient. As a result, the organization becomes much more productive in completing tasks that used to take days or weeks in a much shorter time. This makes the work of ALK members easier and improves the overall efficiency and speed of the organization.

The discussion revolves around creating a user-friendly system interface for clerks to handle applications smoothly. This interface allows clerks to easily access applicant details and payment records, and to update application statuses promptly. A key task for clerks is to review submitted forms, ensuring they meet the requirements set by KADA. With the system's filtering features, clerks can quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. By diligently updating database records, clerks maintain accurate information for organizational use. This streamlined process helps clerks efficiently manage applications, contributing to the organization's overall effectiveness.

5.0 Feasibility

1) Technical

After analyzing the current system, where the applicant has to get and submit the form manually, we found that KADA has the potential to develop an online system. This is because currently all the work is done manually and there are a lot of problems occurring. Therefore, by having an online system, it could increase the work efficiency and the number of customers. To make this a success, KADA has to hire a staff member in order to build and maintain the system. For example, KADA could hire an administrator that is a programmer where he can be a developer, and at the same time he also could be a system tester and expert.

2) Economic

In order to make the system, we have to take into consideration the economic feasibility such as time, cost of doing a full system, cost of business employee time, and estimated cost of software, or software development. To have a clarity of the cost calculation, this is where we conducted a Cost-Benefit Analysis (CBA). By having the CBA, we could analyze and estimate whether the investment is worth it or not for a long term. If it is worth it, then the system is economically feasible, and the project should be proceeding further.

3) Operational

Operational feasibility involves determining whether the system will function and be used after it is implemented and heavily depends on the human resources available for the project. This is why we have to ask the users and our client opinion regarding the current system, whether it needs an improvement or not. We also have to take into consideration the staff condition whether they are okay with the changes that want to be made or not. From the interview with our client, KADA in the past few days, our client wants to make an improvement by making their registration form digitally so that KADA could work more efficiently and increase the number of users in applying for membership and loan.

Cost-benefit analysis

Cost	Year 0	Year 1	Year 2	Year 3
Development cost				
Hardware	22 000			
Software	16 500			
Training	11 000			
Consultation	11 000			
Total development cost	60 500			
Production cost				
IS Support		11 000	11 550	12 128
Supplies		1320	1386	1455
Maintenance		2200	2310	2426
Advertisement		1375	1444	1516
Total production cost		15 895	16 690	17 525
Present value		13 821	12 620	11 523
Accumulated cost		74 321	86 941	98 464
Benefit				
Saving		36 000	38 520	41 216
Increase sale		18 000	19 260	20 608
Total benefit		54 000	57 780	61 824
Accumulated benefit		54 000	111 780	173 604
Gain or lose (npv)		(20 321)	24 893	75 140
Profitability index	1.24			

Cost	
Hardware	20 000
Software	15 000
Training	10 000
Consultation	10 000
IS Support	10 000
Supplies	1200
Maintenance	2000
Advertisement	1250

Benefits	
Savings	40 000
Increase sales	20 000

Discount rate	15%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefit)	0.9
Annual change in cost	5%
Annual change in benefit	7%

6.0 Objectives

1) Simplify the registration process to reduce time and effort

By shifting to a computerized system, many inefficiencies of the current manual system, such as redundant paperwork, slow processing times, and frequent errors, can be greatly reduced or even eliminated. Manual data entry is often slow and error-prone, leading to delays and mistakes. Paper records can be lost or damaged, making data retrieval difficult. A computerized system allows for instant processing and updating of information, reducing these delays. Automated data entry ensures accuracy and consistency. This not only improves efficiency but also frees up employees to focus on more important tasks, enhancing overall productivity and customer satisfaction.

2) Ensure data integrity and security, preventing any loss of information.

Shifting from a manual to a computerized registration system can significantly avoid data loss by implementing data backup, and access control mechanisms. Unlike manual systems, which are prone to physical damage, misplacement, and human error, a computerized system ensures that all data is securely stored in a centralized database with regular automatic backups, reducing the risk of accidental loss.

3) Make the registration process user-friendly and intuitive for all users.

By implementing online forms with real-time data validation and clear guidance, users can complete registrations quickly and accurately. The system's responsive design ensures accessibility across various devices, while features like FAQs, and live support provide immediate assistance when needed. Additionally, automated workflows reduce wait times and streamline operations, resulting in a seamless and efficient experience that enhances user satisfaction and engagement.

4) Enhance profitability by reducing costs and increasing productivity.

Transitioning from a manual to a computerized registration system can significantly increase company profit by reducing operational costs and attracting more new applicators. Automating the registration process minimizes the need for manual labor and physical materials, leading to cost savings. Additionally, the system can handle a higher volume of registrations efficiently and support business growth. Also, it could enhance data accuracy and security and reduce the risk of costly errors and compliance issues.

7.0 Scope and Limitations

The system provides various features. It allows easy membership registration, updates on cooperative events, and loan requests. Members can also update their personal information to ensure accuracy. Cooperative administrators can also efficiently manage member details, monitor finances, and plan events using a centralized interface.

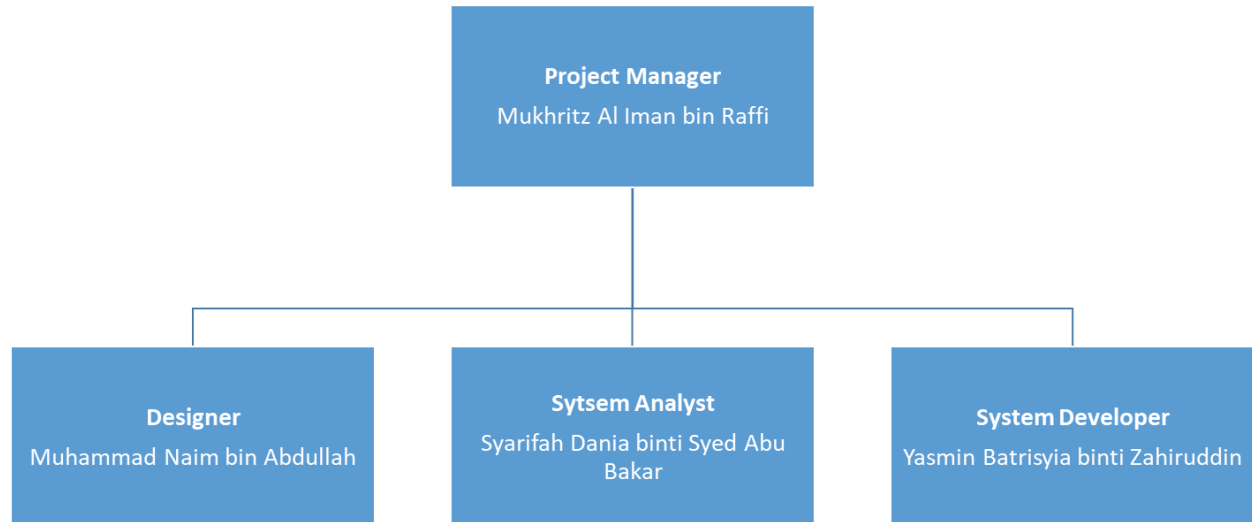
However, our system does have some limitations that we need to address. While it helps with the loan application process, we rely on external financial institutions for the actual disbursement of funds. Moreover, as our membership and loan requests grow, there may be concerns about scalability which could lead to a decrease in the system's performance. Additionally, we must ensure that our system complies with regulatory frameworks related to data privacy and financial transactions, which might restrict its functionalities and require strict adherence.

Despite these challenges, it's important to recognize the value that our proposed system brings to streamlining club management processes. By understanding both its capabilities and limitations, stakeholders can effectively utilize the system while managing potential issues. Regular evaluation and improvement will enable our system to continue enhancing efficiency and effectiveness in club management practices.

In conclusion, our proposed system represents a significant urging in club management technology, offering a robust framework for optimizing membership processes and loan facilitation. By navigating its capabilities and limitations judiciously, organizations can harness its potential to cultivate a thriving club polity while complying to regulatory compliance standards.

8.0 PROJECT PLANNING

8.1 Human Resource

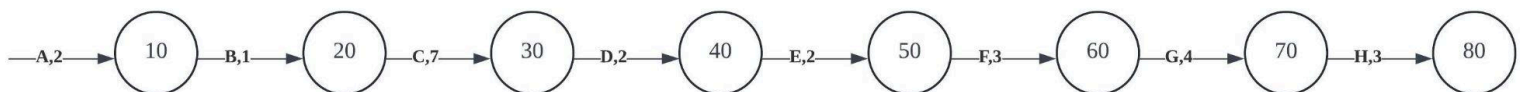


8.2 Work Based Structure



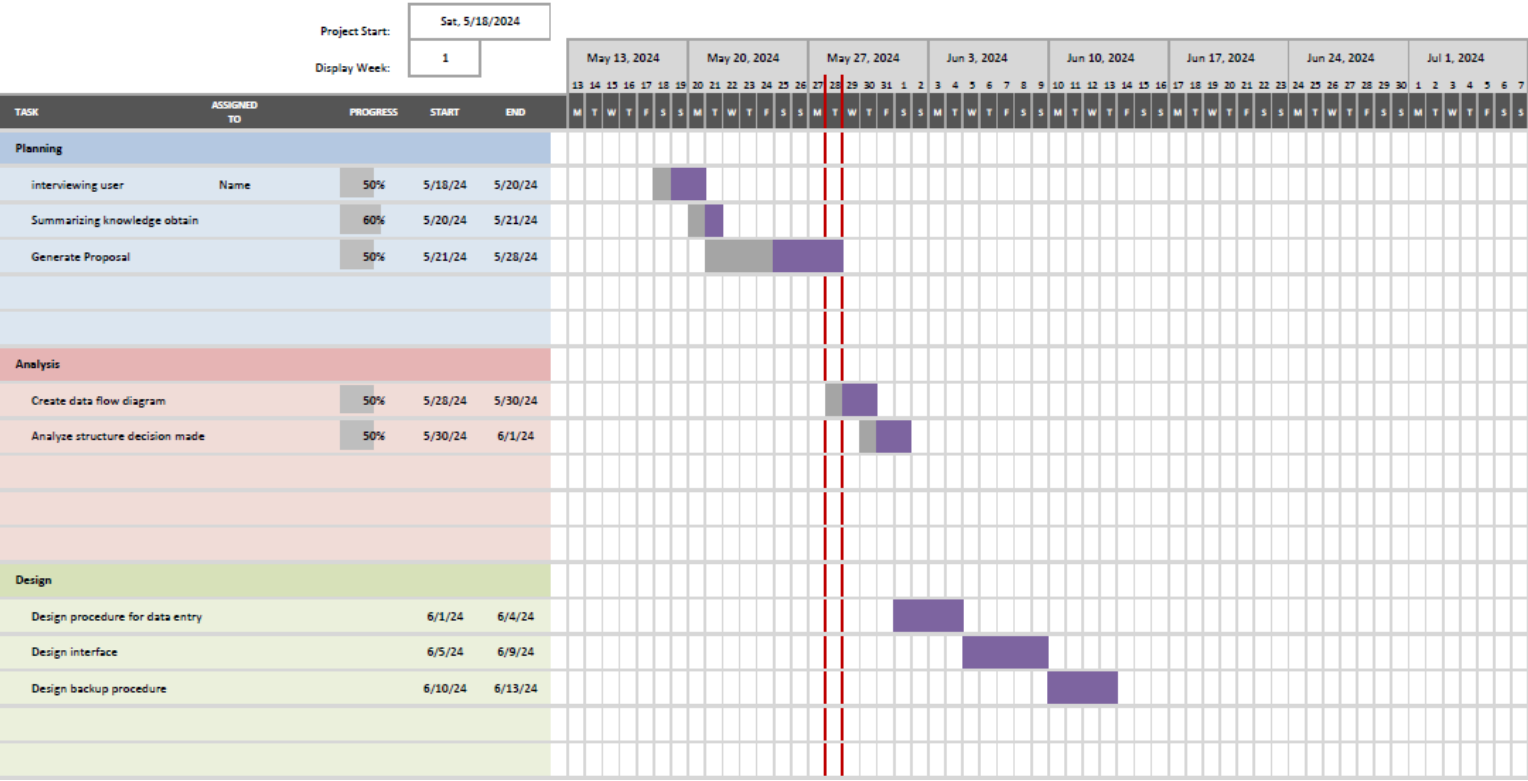
8.3 Pert Chart

Task	Activity	Predecessor	Duration (days)
A	Interviewing user	none	2
B	Summarizing knowledge obtain	A	1
C	Generate proposal	B	7.
D	Create data flow diagram	C	2.
E	Analyze the structure decision made	D	2
F	Design procedure for data entry	E	3
G	Design interface	F	4
H	Design backup procedure	G	3



8.4 Gantt Chart

KADA online registration and loan application



9.0 Requirement Analysis

9.1 Current business process (scenarios, workflow)

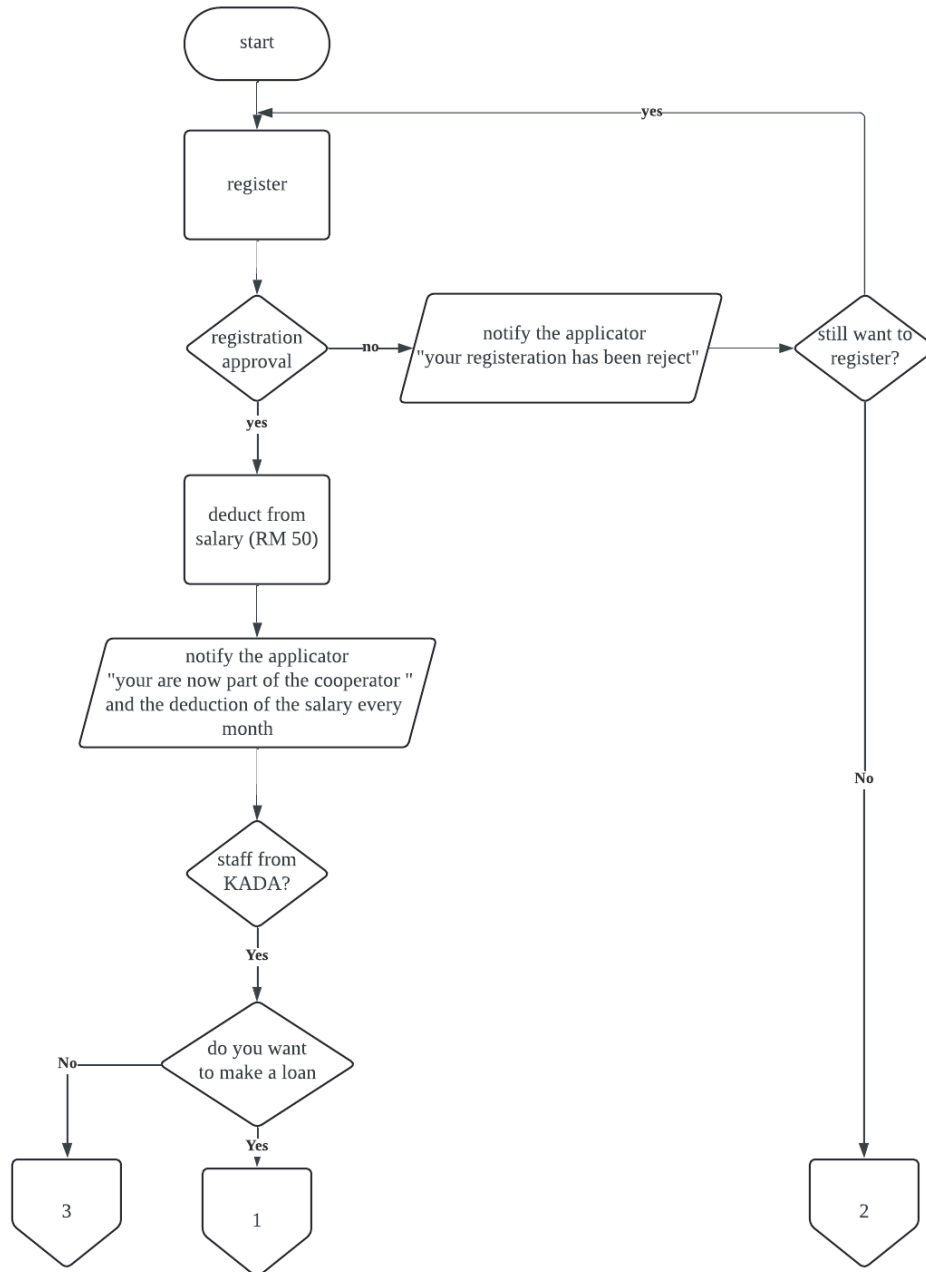


Figure 9.1.1 flowchart of the workflow

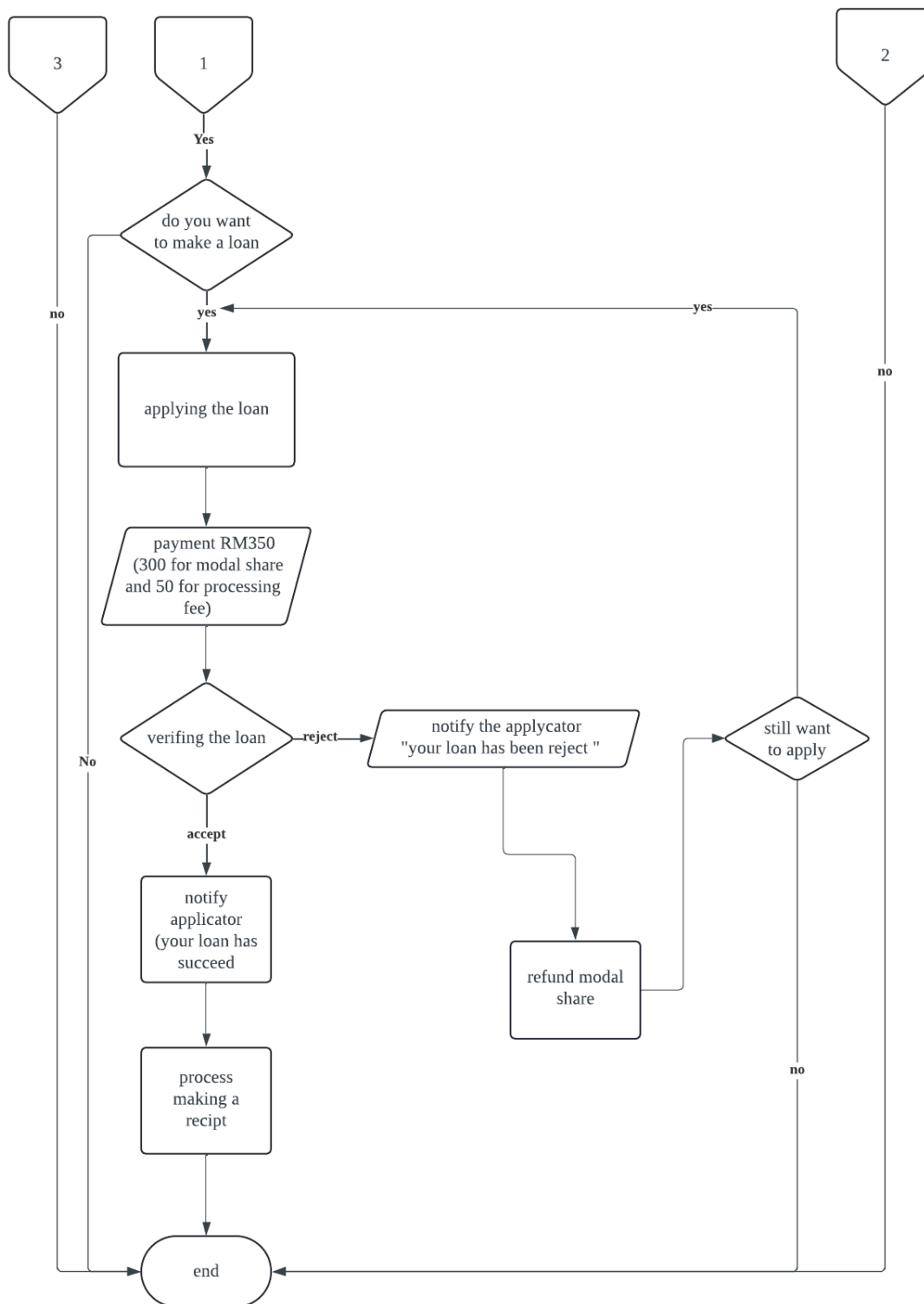


Figure 9.1.2 flowchart of the workflow

9.2 Functional Requirement (input, process and output)

User Registration:

- Users should be able to register for a KADA membership online by filling out and submitting a registration form.
- The system should validate the information entered (e.g., required fields, correct data formats) before submission.
- Users should receive an email confirmation upon successful registration.

Membership Management:

- Users should be able to view and update their membership details.
- Admins should be able to approve or reject membership applications.
- The system should notify users via email about the status of their membership application.

Loan Application:

- Members should be able to apply for a loan online by filling out and submitting a loan application form.
- The system should validate the information entered (e.g., required fields, correct data formats) before submission.
- Users should receive an email confirmation upon successful loan application submission.

Loan Management

- Admins should be able to review, approve, or reject loan applications.
- The system should provide a dashboard for admins to view pending, approved, and rejected loan applications.
- Users should be notified via email about the status of their loan application.

Document Upload:

- Users should be able to upload necessary documents (e.g., proof of identity, income statements) as part of the registration and loan application processes.
- The system should allow admins to view and download these documents for verification.

Notifications:

- The system should send email notifications for important events, such as registration confirmation, loan application status updates, and membership approvals.

User Dashboard:

- Users should have access to a dashboard where they can view their membership status, loan application status, and any notifications.

Admin Dashboard:

- Admins should have access to a dashboard where they can manage user registrations and loan applications.
- The dashboard should provide summary statistics and detailed views of all applications and memberships.

Reporting:

- The system should generate reports on user registrations, loan applications, and other key metrics.
- Admins should be able to download these reports in common formats (e.g., PDF, Excel).

These functional requirements ensure that KADA's online system will support essential processes such as user registration, membership management, loan application processing, and administrative oversight, while providing a secure and user-friendly experience.

9.3 Non-functional Requirement (performance and control)

Performance:

- The system shall handle up to 2000 simultaneous users without significant performance issue.

Usability:

- The system will provide a user-friendly interface that is easy to navigate.

Security:

- The system will employ security measures such as firewall to prevent unauthorized access

Maintainability:

- The system shall ensure a straightforward system design that allows for easy updates and maintenance

Backup and Recovery:

- The system will implement regular backups of critical data.

9.4 Logical DFD AS-IS system (Context Diagram, Diagram 0)

CONTEXT DIAGRAM

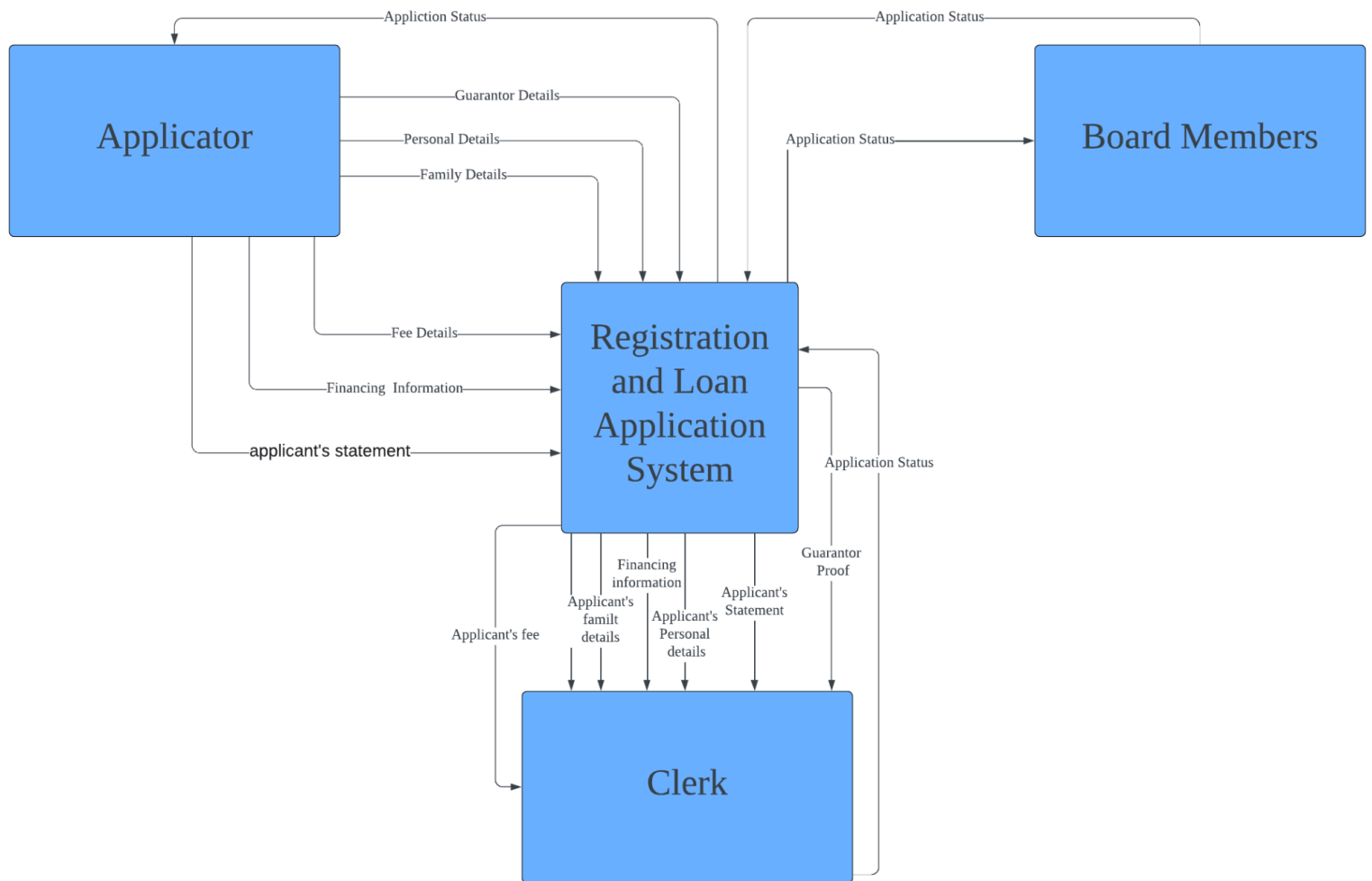
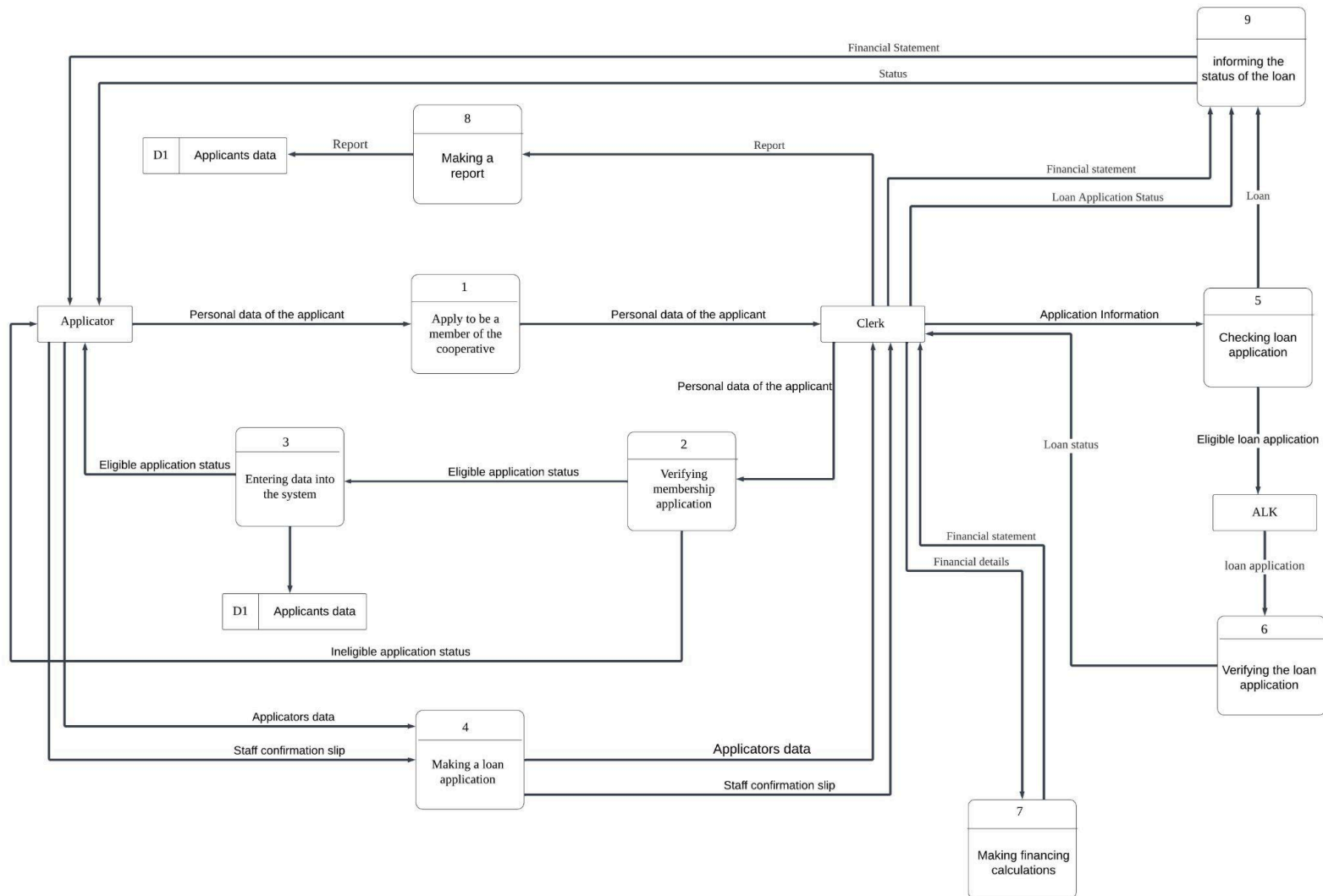


DIAGRAM 0



10.0 Benefit and Overall Summary of Proposed System

The proposed system is set to significantly enhance the efficiency and productivity of KADA by transitioning from a manual, paperwork-based system to a modern online platform. This new system will continue to perform the essential tasks of handling registration forms for new members and loan applications but will do so in a more streamlined and efficient manner. By moving these processes online, the system introduces numerous benefits that cater to the needs of applicants, ALK members, clerks, and the organization as a whole.

For applicants, the online system offers unmatched ease of use and availability. Users no longer need to make physical trips to the headquarters to pick up and submit forms, which can be particularly time-consuming and costly, especially for those living far away or with busy schedules. The ability to access and submit forms online at any time from anywhere with an internet connection makes the process significantly easier and more flexible. Moreover, the system provides instant confirmation of submissions, alleviating any concerns about whether forms have been received by KADA. This feature helps reduce errors and ensures that applications are completed accurately and efficiently.

The new system also brings substantial benefits to the members of the Board of Directors (ALK). Previously, their task of reviewing membership and loan applications was hindered by the need to be physically present in the office and manage numerous paper documents. The online system removes these barriers, allowing ALK members to perform their duties remotely, thus enhancing their flexibility and responsiveness. This remote capability means applications can be reviewed and approved more quickly, leading to faster decision-making processes. The system's instant updates and notifications further streamline their workflow, making the entire process more efficient and productive.

Clerks also stand to gain significantly from the proposed system. With a user-friendly interface, clerks can access applicant details and payment records with ease, and promptly update application statuses. The system's filtering features enable clerks to quickly sort through applications, accepting those that meet criteria and rejecting incomplete or non-compliant ones. This efficiency in handling applications reduces the manual workload and helps maintain accurate and up-to-date database records. As a result, clerks can manage applications more effectively, contributing to the overall effectiveness of the organization.

Overall, the transition to an online system represents a major upgrade for KADA. It not only improves the accuracy and security of data management by reducing the risk of handling sensitive personal information manually but also enhances the speed and efficiency of service delivery. By adopting this digital solution, KADA will experience higher productivity and greater user satisfaction, positioning itself as a more efficient and responsive organization. The proposed system is thus a powerful and effective substitute for the outdated manual processes, bringing significant improvements across all levels of operation.